

MODULE 1: Management Studio

Exercise 1. Creating a link

Task:

- Default: start, duration, predecessor
- Summary:
- Recurring Tasks:
- Milestone:

Creating the following tasks

ID	TASK NAME	PREDECESSOR	DURATION
1	Research destinations	-	2 days
2	Book flights	1	1 day
3	Prepare and pack	2	5 days
4	Fly out	3	1 day
5	Relax in the sun	4	7 days
6	Fly home	5	1 day
7	Peel	6	7 days

- Display the Project Information
- View Project Timeline, Gant Chart, Network Diagram and capture the result them.

Exercise 2. Setting duration

Creating the following tasks

ID	TASK NAME	PREDECESSOR	DURATION
1	Quarry stone	-	2 days
2	Transport stone	1	1 days
3	Prepare stone	2	5 days
4	Fill foundations	1,3	1 day
5	Level ground	2,3	1 day
6	Measure site	1	7 days
7	Mark out site	5	5 days
8	Acquire stone	2,4	3 days
9	Prepare site	5,3	9 days
10	Build	8,9	2 days

- Display the Project Information
- View Project Timeline, Gant Chart, Network Diagram and capture the result of them.

Exercise 3. Creating phases

Creating the following tasks

ID	TASK NAME	PREDECESSOR	DURATION
1	Pre-heat oven	-	5 mins
2	Prepare meat	1	5 mins
3	Prepare veg	2	5 mins
4	Heat pan	3	5 mins
5	Fry veg	4	5 mins
6	Brown meat	5	5 mins

7	Add stock and bring to a simmer	6	5 mins
8	Cook in oven	7	45 mins

- Group the eight tasks so that the first task becomes a summary task (Make a casserole 1 day?)
- Display the Project Information
- View Project Timeline, Gant Chart, Network Diagram and capture the result of them.

Exercise 4. Noah - Linking

Create the following links

ID	TASK NAME	PREDECESSOR	DURATION
1	Build boat	-	7 days
2	Load supplies	1	3 days
3	Load family	1	6 days
4	Load animals	1	7 days
5	Collect animals A-M	1,4	12 days
6	Collect animals N-Z	2,4	10 days
7	Float around	5,6	7 days

- Display the Project Information
- View Project Timeline, Gant Chart, Network Diagram and capture the result of them.

Exercise 5. Pyramid - Multi level grouping

Create the following links

ID	TASK NAME	PREDECESSOR	DURATION
1	<i>Build a pyramid</i>	-	<i>1 day?</i>
2	<i>Acquire stone</i>	1	<i>1 day?</i>
3	Quarry stone	2	3 wks
4	Transport stone	3	5 days
5	Prepare stone	4	3 wks
6	<i>Prepare site</i>	5	<i>1 day?</i>
7	Dig foundations	6	3 wks
8	Fill foundations	7	2wks
9	Level ground	8	1wk
10	Measure site	9	7 days
11	Mark out site	10	7 days
12	Build	11	6 mons

- Group the tasks as indicated above with 3 levels:
 - Level 1 at task 1 includes task 2,6,12
 - Level 2 at task 2 includes task 3,4,5, task 6 includes task 7,8,9,10,11
- Display the Project Information
- View Project Timeline, Gant Chart, Network Diagram and capture the result of them.

Exercise 6. Creating a plan for Staff Training

ID	TASK NAME	PREDECESSOR	DURATION
1	Start with a skills gap analysis.	-	7 days

2	Identify employees for career	1	3 days
3	Align with employee and company goals.	2	6 days
4	Help them grow with you	3	7 days
5	<i>Use employee development plan</i>	4	1 days?
6	Consider an individual employee	5	2 wks
7	A succession planning template	6	1wks
8	<i>Fit the learning opportunity to the training</i>	7	1 days?
9	Microlearning	8	2 wks
10	On-the-job training	9	5wks
11	Augmented reality	10	1wks
12	Track results to inform your decisions	11	7 days

- Group the tasks as indicated above with 3 levels:
 - Level 1 at task 1,2,3,4,5,8,12.
 - Level 2 at task 5 includes task 6,7 and task 8 includes task 9,10,11
- Display the Project Information
- View Project Timeline, Gant Chart, Network Diagram and capture the result of them.

Exercise 7. Recurring tasks

Create plan for a software project

ID	TASK NAME	PREDECESSOR	DURATION
1	Planning	-	1 wk
2	Requirement documents	1	2 wks
3	Sign off	2	1 day
4	Development	3	10 wks
5	Testing	4	4 wks
6	Implementation	5	2 wks
7	Review	6	1 wk

- Add in a weekly meeting to run throughout the project with the following settings
 - Name: Breakfast Meeting
 - Duration: 1 hour
 - Recurrence: Weekly
 - Day: Monday
 - Range: the start to end of the project.
 - Calendar: Standard
- Display the Project Information
- View Project Timeline, Gantt Chart, Network Diagram and capture the result of them.

Exercise 8. Travelling – task calendar

You intend to travel from HCM to Kuala Lumpur, Bangkok and then, come back to HCM. Creating a plan for your trip.

MODULE 2: CALENDARS AND SCHEDULING

Exercise 1. Part Time - creating new calendar

Open project file Ex2 (Module 1), Use the **Project Information** dialog box to:

- A part-time worker will be used on this project. Go to the **Change Working Time** dialog box and create a new copy of the **Standard** calendar, calling it "Part Time".
- Update the working time of your new calendar so that:
 - Mondays to Fridays are nonworking time
 - The remaining 2 weekdays have working time of 10:00 to 16:00 (no breaks)
- The Part time has also booked a holiday from 1st to Feb 5th. Create an **Exception** to their calendar with a **Holiday** name and these days as nonworking
- Display the Project Information.

Exercise 2. Changing Working Time

Open project file Ex4 (Module 1) Use the **Project Information** dialog box to:

- Start Date: Nov 20th 2025
- Go to the **Change Working Time** dialog box and create an **Exception** called "Tet Holidays" during from Dec 24th 2026 to Jan 5th 2026 into nonworking time.
- Change **Work Weeks time** of **Standard** calendar's default working week. This project's standard working time are:
 - 08:30 to 17:30 Monday to Thursday (no lunch break)
 - 08:30 to 16:00 on Fridays (no lunch break)
 - Add a new Work Week to the **Standard** calendar named "**Xmas Slowdown**". Use this to model a 12:30 finish on every weekday from Dec 10th 2025 to Dec 23rd 2025.
 - Capture the Work weeks time

Exercise 3. Changing Working Time

Open project file Ex5 (Module 1), Use the **Project Information** dialog box to:

- Star Date: Apr 5th 2026
- The office will shut down for 2 weeks in the summer. Use the **Change Working Time** dialog box to create an **Exception** called "**Summer Shutdown**" and make the first 14 workdays of **August** nonworking time
- Create an Exception called "Independence Days" is 2 Sep nonworking time.
- Create an Exception called "International Labor days" is Apr 30th and May 1st nonworking time
- Display the Project Information

Exercise 4. Changing Working Time

Open project file Ex6 (Module 1), Use the **Project Information** dialog box to:

- Modify the **Standard** calendar and change the **[Default]** work week to use the hours shown below:

Details for '[Default]'

Set working time for this work week

Select day(s):

- ☐ Use Project default times for these days.
- ☐ Set days to nonworking time.
- ☒ Set day(s) to these specific working times:

	From	To
1	8:00 SA	12:00 CH
2	1:00 CH	5:00 CH

Buttons: Help, OK, Cancel

- Add an exception in the **Standard** calendar called **Recovery** with in 7 lates workdays of March is non-working days.
- Display the Project Information.

Exercise 5. Create the private Time for a project

Open project file Ex7 (Module 1), Use the **Project Information** dialog box to:

- Create the new Time Work for your project. [NameTimeProject]
{Name: Your Name - Ex: MinhTimeProject}
- Change **Work Weeks time** of **Standard** calendar's default working week. This project's standard working time are:
 - 08:00 to 12:00 on Mon, Wed, Fri
 - 13:00 to 16:00 on Tue, Thu, Sat
 - Create an Exception called "International Labor days" is Apr 30th and May 1st nonworking time
 - Capture the Work weeks time
- Display the Project Information.

Exercise 6. Your project

- Create your plan from Monday to Sunday. It will list all of your activities, your time you spend and resources joining in each task.
- Create a new calendar called **Student** based on the **Standard** calendar.
- Change the **[Default]** work week so that the hours match to your calendar everyday
- Using Note function in MP 2010 to take note for important tasks in your plan.

MODULE 3: Working with Tasks

Exercise 1. Changing Working Time (Start Date Feb 2st 2026)

- Santa's workshop has two machines for making toys for boys and girls - the first job of the year is to prepare them for work.

ID	TASK NAME	PREDECESSOR	DURATION
1	clean machine		5 days
2	load raw materials	1	3 days
3	prime machine	2	3 days
4	Test run	3	1 day
5	clean machine	4	5 days
6	load raw materials	5	3 days
7	prime machine	6	3 days
8	Test run	7	1 day

Create the summary tasks for series tasks below:

- Summary task name is **Boy's Toy** for tasks 1,2,3,4
- Summary task name is **Girl's Toy** for tasks 5,6,7,8
- Add the **Ready for production task** (0 day) that is the end task of the task list
- Summary Task name is **Workshop** for **Boy's Toy**, **Girl's Toy**, and **Ready for production tasks**
- Display the Project Information.
- View Project Timeline, Gantt Chart, view CP, Total Slack and capture the result of them

Exercise 2. Changing Working Time (Start Date Feb 2nd 2026)

- Using the diagram below, create a list of tasks to model this process.

ID	TASK NAME	PREDECESSOR	DURATION
1	Visit estate agents		1 days
2	View properties	1	2 days
3	Decide what to buy	2	5 days
4	Make offer	3	0 day
5	Wait for decision	4	<i>Don't know</i>
6	offer a accepted	5	0 days

- Create the summary task is Buy a shop that contains all tasks in the project.
- Establish the duration of **Wait for a decision task** is 7 days.
- Change the duration of **View properties task** to **7 days**.
- Display the Project Information.
- View Project Timeline, Gantt Chart, view CP, Total Slack and capture the result of them.

Exercise 3. Scheduling – Task dependencies

- Create the list of tasks to model this processing (Start date: Feb 2nd 2026)

ID	TASK NAME	Duration
1	Creating architectural plans	3 wks
2	Submit plans for approval	1 mon
3	Order materials	8 days
4	Erect fencing	3 days
5	Erect site building	4 days
6	Clear and level site	3 wks
7	Prepare drainage infrastructure	1 wk
8	Prepare cabling infrastructure	1 wk
9	Pour foundations	4 days
10	Erect steelwork	3 mons
11	Erect wall	2 mons
12	Install roofing superstructure	2 wks
13	Install roofing retracting mechanism	1 wk
14	Erect seating tiers	3 wks
15	Fit all windows and doors	2 wks
16	Install electrical cabling	1 wk
17	Install electrical fittings and fixtures	2 wks
18	Install all plumbing	2 wks
19	Install plumbing fixtures and fittings	1 wk
20	Lay astro turf	1 wk
21	Erect handrails and fencing	2 wks
22	Paint rooms, fixtures, fittings, etc.	1 mon
23	Install PA system	2 days
24	Install video imaging equipment	3 days
25	Fit out control room	1 wk
26	Test roof mechanism	1 wk
27	Test PA system	2 days
28	Test video imaging equipment	2 days
29	Test control room equipment	1 wk
30	Obtain official occupancy	1 day

31	Obtain safety certification	2 days
32	Official opening	1 day

Create the summary tasks for series tasks below:

- Summary task name is **Planning (20 days)** for tasks 1,2,3.
- Summary task name is **Site Works (15 days)** form task 4 to task 8.
- Summary task name is **Building Construction (60 days)** form task 9 to task 14.
- Summary task name is **Fit Out (20 days)** form task 15 to task 25.
- Summary task name is **Commissioning (5 days)** form task 26 to the end task.

Create the milestones for this project below:

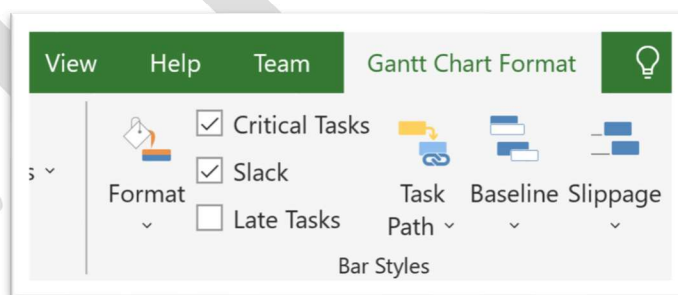
No	Milestone	At End Of phrases
1	Planning Completed	<i>Planning</i>
2	Site Works Completed	<i>Site works</i>
3	Building Construction Completed	<i>Building Construction</i>
4	Fit Out Completed	<i>Fit out</i>
5	Commissioning Completed	<i>Commissioning</i>

Creating Dependencies in Task Entry

ID	TASK NAME	Duration	Dependencies
1	Planning	?????	
2	Creating architectural plans	3 wks	
3	Submit plans for approval	1 mon	2
4	Order materials	8 days	3
5	Planning Completed	0	2,3,4
6	Site works		
7	Erect fencing	3 days	5
8	Erect site building	4 days	7
9	Clear and level site	3 wks	8
10	Prepare drainage infrastructure	1 wk	9
11	Prepare cabling infrastructure	1 wk	9
12	Site works Completed	0	7,8,9,10,11
13	Building Construction		
14	Pour foundations	4 days	12
15	Erect steelwork	3 mons	14
16	Erect wall	2 mons	15
17	Install roofing superstructure	2 wks	16
18	Install roofing retracting mechanism	1 wk	17
19	Erect seating tiers	3 wks	18SS
20	Building Construction Completed	0	14,15,16,17,18,19
21	Fit Out		
22	Fit all windows and doors	2 wks	20
23	Install electrical cabling	1 wk	20

24	Install electrical fittings and fixtures	2 wks	23
25	Install all plumbing	2 wks	20
26	Install plumbing fixtures and fittings	1 wk	25
27	Lay astro turf	1 wk	20
28	Erect handrails and fencing	2 wks	20
29	Paint rooms, fixtures, fittings, etc.	1 mon	28
30	Install PA system	2 days	23
31	Install video imaging equipment	3 days	23
32	Fit out control room	1 wk	29,30,31
33	Fit out Completed	0	22,23,24,25,26,27,28,29,30,31,32
34	Commissioning		
35	Test roof mechanism	1 wk	33
36	Test PA system	2 days	33
37	Test video imaging equipment	2 days	33
38	Test control room equipment	1 wk	35,36,37
39	Obtain official occupancy	1 day	38
40	Obtain safety certification	2 days	39
41	Official opening	1 day	40
42	Commissioning Completed	0	35,36,37,38,39,40,41

- Display the Project Information.
- View the project in Gantt Chart and define CP and Project Slack
- Choose **Gantt Chart Format** tab → check on **Critical Tasks** in **Bar Styles**.



- Check on **Slack** in **Bar Styles** slack lines in non-critical tasks
- On the table data Add column **Late Start**, **Late Finish**, **Total Slack**

MODULE 4: Assigning resources to the tasks in Project

There are three types of resources used to manage and allocate project work effectively. These resources help in scheduling, budgeting, and tracking progress.

Resource Name	Type	Material	Initials	Group	Max.	Std. Rate	Ovt.	Cost/Use	Accrue	Base	Code
Peter - Developer	Work		P		100%	\$10.00/hr	\$0.00/hr	\$1.00	Prorated	Standard	
Mary - Tester	Work		M		100%	\$20.00/hr	\$0.00/hr	\$2.00	Prorated	Standard	
Hard drive	Material	pack	H			\$70.00		\$0.00	Prorated		
Cloud Service	Material	service	C			\$10.00		\$0.00	Prorated		
Travel Expenses	Cost		T						Prorated		
Legal & Compliance Costs	Cost		L						Prorated		

1. Work Resources (People & Equipment)

Definition:

- These are resources that perform work on tasks in the project.
- Their tasks is measured in units of time (hours, days, weeks).
- They can have a defined capacity (% availability), work schedules, and **hourly/daily/monthly/yearly costs**. *[Std.Rate], [Cost/Usage]*
- They can get a private working time *[Base]*

Examples:

- **People:** Software Developers, System Architects, Business Analysts, Project Managers, QA Testers, UI/UX Designers, Database Administrators, Network Engineers, Cybersecurity Experts.
- **Equipment:** Servers, Cloud Computing (e.g., AWS, Azure), Version Control Systems – GitHub, GitLab, or Bitbucket for managing code versions.

How it works in Microsoft Project 2016:

- When a work resource is assigned to a task, Microsoft Project calculates the amount of work based on the resource's availability and the task duration.
- You can set work schedules to ensure proper workload distribution and avoid over-allocation.

2. Material Resources (Consumable Items)

Definition:

- These are physical materials that are consumed as work progresses.
- They do not have a work capacity (availability % does not apply).
- They are measured in units like **GB, TB, units, meters, liters, kilograms, or pieces** instead of time.

Examples:

Hardware & Consumable IT Equipment

- Storage Devices – Hard drives, SSDs, USB flash drives.
- Cables & Networking Accessories – Ethernet cables, adapters, routers.
- Backup Media – DVDs, tapes, external hard drives for data backups.
- Batteries & Power Supplies – UPS batteries, laptop chargers.

Software Licenses & Subscriptions

- Operating System Licenses – Windows, macOS, Linux distributions.
- Development Tools – Visual Studio, JetBrains IDEs, MATLAB.
- Database Licenses – SQL Server, Oracle, MySQL Enterprise.
- Cloud Services – AWS, Azure, Google Cloud storage and computing.

Office & Miscellaneous Supplies

- Printed Documents & Manuals – Requirement specifications, test plans.
- Stationery – Whiteboards, markers, sticky notes for agile teams.
- Headsets & Microphones – Used for remote meetings and testing.

How it works in Microsoft Project 2016:

- When adding material resources, you specify the unit of measurement (e.g., GB, TB, meters, units, licenses, pages, packs, ...).
- Microsoft Project does not schedule material resources based on work time but tracks the quantity required for each task.

3. Cost Resources (Fixed Expenses)

Definition:

- These represent financial costs that are associated with a task but are not dependent on work time.
- Unlike work resources, cost resources do not have a fixed availability or work rate.

Examples:

Software & Licensing Costs

- Cloud Services Subscription – AWS, Azure, Google Cloud fees.
- Software Licenses – Microsoft Office, Adobe Suite, Visual Studio.

- Database Licensing Fees – SQL Server, Oracle, MySQL Enterprise.

Infrastructure & Hardware Costs

- Server Rental – Monthly or yearly hosting fees.
- Data Center Costs – Fees for colocation or cloud hosting.
- Networking Costs – ISP charges, VPN subscriptions.

Professional & Outsourcing Fees

- Consulting Fees – Payments to external consultants.
- Freelancer/Contractor Payments – Web developers, security experts.
- Training & Certification Costs – Courses for IT staff.

Business & Operational Costs

- Travel Expenses – Flights, hotels, transportation for meetings.
- Conference Fees – IT summits, training workshops.
- Legal & Compliance Costs – GDPR, software patents, security audits.

How it works in Microsoft Project 2016:

- When assigning a cost resource to a task, you manually enter the amount.
- Microsoft Project does not calculate cost resources based on time worked; instead, it simply adds them to the task budget.

Exercise 1. Entering Resources

- Open the Exercise 1 (Module 3)
- Tab View → Resource Sheet / Tab Task → Resource Sheet

Add the resources for this project bellow: Work (the person)

STT	Name	Initials	Groups	Max. Units	Std. Rate
1.	David	D	Consultant	100%	8
2.	Brian	B	Staff	100%	7
3.	Mary	M	Staff	100%	7
4.	John	J	Staff	100%	7
5.	Worker 1: Liam	W1	Worker	100%	6
6.	Worker 2: Chang	W2	Worker	100%	6

- Assign the resources for each task in the project

Tab Task → Gantt Chart → Add Column Resource Name, Cost

Select the task that you want to assign the resources:

ID	TASK NAME	Resource Names	Cost
1	clean machine	Worker 1, David	
2	load raw materials	Worker 2	
3	prime machine	Mary	
4	Test run	Brian	
5	clean machine	Worker 1	
6	load raw materials	Worker 2	
7	prime machine	John	
8	Test run	David	

Capture images:

- Gantt Chart (Resource Name, Cost Column)
- Display the Project Information (Statics): Duration, Hours, Cost
- Resource Sheet
- Resource Usage

Exercise 2. Entering resources

- Open the Exercise 2 (Module 3)
- Tab View → Resource Sheet

Add the resource for this project bellow: Material Type

STT	Name	Type	Materials	Initials	Groups	Std. Rate
1.	MDF	Material	Sheet	Md	Material	20
2.	Flooring	Material	Pack	Fl	Material	80
3.	Purple Material	Material	Roll	Pu	Material	30
4.	Orange Material	Material	Roll	Or	Material	120

- Assign the resources for each task in the project

Tab Task → Gantt Chart → Add Column Resource Name, Cost

The information of resources that assigning to each task

ID	TASK NAME	RESOURCE Names	Cost
1	Visit estate agents	MDF [2]	
2	View properties	Flooring [2]	
3	Decide what to buy	Purple Material [1]	
4	Make offer	Flooring [1], Purple Material [1]	
5	Wait for decision	Flooring [1], Orange Material [1]	
6	offer a accepted	Orange Material [2]	

Capture images:

- Gantt Chart (Resource Name, Cost Column)
- Display the Project Information (Statics): Duration, Hours, Cost
- Resource Sheet
- Resource Usage

Exercise 3. Entering resources

- Create the project bellowing:

ID	TASK NAME	PREDECESSOR	DURATION
1	Planning	-	1 wk
2	Requirement documents	1	2 wks
3	Sign off	2	1 day
4	Development	3	10 wks
5	Testing	4	4 wks
6	Implementation	5	2 wks
7	Review	6	1 wk

- Tab View → Resource Sheet / Tab Task → Resource Sheet

Add the resources this project bellow: Work/Material

STT	Name	Type	Materials	Initials	Groups	Std. Rate
1.	PM	Work		PM	Sponsor	10
2.	Designer	Work		Ds	Staff	6
3.	Developer	Work		Dp	Staff	8
4.	Tester	Work		Tt	Staff	6
5.	Maintainer	Work		M	Staff	5
6.	Operating System	Material	Unit	OS	Material	120
7.	Internet Provider	Material	Unit	ISP	Material	100
8.	Cable	Material	M	Cb	Material	1
9.	Visual Studio	Material	Unit	VS	Material	20

- Assign the resource for each task in the project

ID	TASK NAME	Work	Material
1	Planning	PM	
2	Requirement documents	PM, Designer	
3	Sign off	PM	
4	Development	Developer	Operating System[1] , Visual Studio [1]
5	Testing	Tester	
6	Implementation	Maintainer	Operating System , Visual Studio, Cable [100]
7	Review	PM, Maintainer	

Capture images:

- Gantt Chart (Resource Name, Cost Column)
- Display the Project Information (Statics): Duration, Hours, Cost
- Resource Sheet
- Resource Usage

Exercise 4. Entering resources

- Open the Exercise 3 (Module 3)
- Tab View → Resource Sheet → Resource Views

Add the resources that type is work (default) for the project below:

STT	Name	Initials	Groups	Max. Units	Std. Rate
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1.	Architect	Arc	Consultant	100%	9
2.	Draftsperson	Dft	Staff	200%	7
3.	Building Clerk	BC	Staff	100%	8
4.	Supervisor	Sup	Staff	100%	12
5.	Rigger	Rig	Wages	600%	5
6.	Boilermaker	BM	Wages	600%	5
7.	Welder	Weld	Wages	500%	6
8.	Carpenter	Car	Wages	800%	7
9.	Painter	Ptr	Wages	500%	6
10.	Labourer	Lab	Wages	1,000%	6
11.	Driver	Drv	Wages	300%	8
12.	No Barrier Fencing	NBF	Contractor	100%	3
13.	Rock Solid Concrete	RSC	Contractor	100%	4
14.	Listen Ear Audio	LEA	Contractor	100%	5
15.	In Focus Video	IFV	Contractor	100%	5
16.	Pure Grass Turf	PGT	Contractor	100%	5
17.	Building Inspector	BI	Government	100%	5
18.	High Jib Crane	HJC	Equipment	100%	7
19.	Grader	Grd	Equipment	200%	6
20.	Air Compressor	AC	Equipment	100%	7
21.	Utility	Ute	Equipment	200%	5
22.	Electrician	Cap	Staff	400%	7
23.	Plumber	Plu	Staff	400%	5

- Establish the resources that type is material for the project below at the of Resource list above:

STT	Name	Type	Materials	Initials	Groups	Std. Rate
24.	Astro Turf	Material	Square Matre	Grass	Material	12
25.	Paint	Material	Ultre	Paint	Ute	20

- Assign the resources for each task in the project

ID	TASK NAME	RESOURCE
1	Planning	
2	Creating architectural plans	Architect, Draftsperson (200%), Building Clerk (25%)
3	Submit plans for approval	Architect
4	Order materials	Draftsperson, Building Clerk
5	Planning Completed	
6	Site works	
7	Erect fencing	No Barrier Fencing
8	Erect site building	Carpenter (400%), Supervisor, Labourer
9	Clear and level site	Supervisor, Grader(200%), Driver(200%)
10	Prepare drainage infrastructure	Supervisor(50%), Plumber(200%)
11	Prepare cabling infrastructure	Supervisor(50%), Electrician(300%)
12	Site works Completed	
13	Building Construction	
14	Pour foundations	Rock Solid Concrete

15	Erect steelwork	Supervisor [50%], Rigger [600%], Boilermaker [600%], Welder [500%], Labourer [600%], Driver [200%], High Jib Crane, Utility
16	Erect wall	Supervisor [50%], Carpenter [700%], Labourer [400%], Driver, Grader, Air Compressor, Electrician [75%], Plumber [25%]
17	Install roofing superstructure	Supervisor [50%], Rigger [500%], Boilermaker [500%], Welder [300%], Labourer [500%], Driver [200%], High Jib Crane, Utility, Air Compressor, Plumber [50%]
18	Install roofing retracting mechanism	Supervisor [50%], Welder, Boilermaker [200%], Rigger [200%], Electrician [200%], Driver, High Jib Crane
19	Erect seating tiers	Supervisor [50%], Carpenter [800%], Welder [200%], Boilermaker [200%], Labourer [500%], Driver, Utility, Air Compressor
20	Building Construction Completed	
21	Fit Out	
22	Fit all windows and doors	Carpenter [500%], Labourer [200%]
23	Install electrical cabling	Electrician [300%], Labourer [200%]
24	Install electrical fittings and fixtures	Electrician [300%]
25	Install all plumbing	Plumber [200%], Labourer [200%]
26	Install plumbing fixtures and fittings	Plumber [200%]
27	Lay astro turf	Pure Grass Turf
28	Erect handrails and fencing	Welder [400%], Boilermaker [200%]
29	Paint rooms, fixtures, fittings, etc.	Painter [500%]
30	Install PA system	Listen Ear Audio
31	Install video imaging equipment	In Focus Video
32	Fit out control room	Electrician [200%], Listen Ear Audio, In Focus Video
33	Fit out Completed	
34	Commissioning	
35	Test roof mechanism	Supervisor, Electrician [200%], Rigger
36	Test PA system	Listen Ear Audio
37	Test video imaging equipment	In Focus Video
38	Test control room equipment	Supervisor, Listen Ear Audio [20%], In Focus Video [20%], Architect, Electrician [200%]
39	Obtain official occupancy	Architect, Building Clerk, Supervisor
40	Obtain safety certification	Supervisor
41	Official opening	Supervisor[200%]

42

Commissioning Completed

Capture images:

- *Gantt Chart (Resource Name, Cost Column)*
- *Display the Project Information (Statics): Duration, Hours, Cost*
- *Resource Sheet*
- *Resource Usage*

Exercise 5. Resources with the private working time

- Creating the Calendars with working times:

PM_Time:

1. Working time: Mon – Sat: 7AM – 11AM
2. Nonworking time: Sun
3. Exception: Nonworking time: 30/4, 1/5, 2/9

Dev_Time:

1. Working time: Mon – Fri: 8AM – 11AM, 1PM – 5PM
2. Nonworking time: Sat, Sun
3. Exception: Nonworking time: 30/4, 1/5, 2/9

Extra_Time:

1. Working time: Mon – Fri: 6PM – 9PM
 2. Nonworking time: Sat, Sun
 3. Exception: Nonworking time: 30/4, 1/5, 2/9
- Creating the following tasks

No	Task Name	Duration (days)	Dependencies
1	Requirement Analysis	5	-
2	Project Planning	3	1
3	UI/UX Design	7	1
4	Database Design	5	1
5	Backend Development	14	4
6	Frontend Development	14	3
7	Frontend & Backend Integration	5	5, 6
8	Testing & Bug Fixing	7	7
9	Deployment to Test Environment	3	8
10	Evaluation & Optimization	5	9
11	Official Deployment	2	10
12	Maintenance & Support	Ongoing	11

- Tab View → Resource Sheet / Tab Task → Resource Sheet
Add the resources this project below: Work/Material/Cost
Each Resource bases on a type of working time:

No.	Name	Type	Quantity	Unit	Std.Rate	Base	Related Tasks
1	Project Manager	Work	1	Person	20	PM_Time	Project Planning

2	System Analyst	Work	3	Person	15	Extra_Time	Requirement Analysis, Database Design
3	Backend Developer	Work	4	Person	18	Dev_Time	Backend development, Frontend & Backend Integration
4	Frontend Developer	Work	3	Person	17	Dev_Time	Frontend development
5	UI/UX Designer	Work	2	Person	16	Extra_Time	UI/UX design
6	Tester	Work	3	Person	16	Dev_Time	Testing & bug fixing
7	Cloud Server (AWS, Azure)	Cost	1	Subscription	200		Data storage, application deployment
8	Software Licenses	Cost	1	Software Package	99		Deployment to Test Environment
9	Developer Workstations	Material	4	Units	1200		Deployment to Test Environment
10	UI/UX Design Tools	Cost	1	Software Package	200		Official Deployment
11	Testing Tools (Selenium, JMeter)	Cost	1	Software Package	90		Testing & bug fixing, Evaluation & Optimization

Capture images:

- Gantt Chart (Resource Name, Cost Column)
- Display the Project Information (Statics): Duration, Hours, Cost
- Resource Sheet
- Resource Usage

Exercise 6. The Engagement planning

Creating the following tasks

STT	Name	Duration	Dependencies
-----	------	----------	--------------

1.	The Engagement	1 day	
2.	The Venue - Confirm Dates	4 wks	1
3.	The Photographer	3 wks	2
4.	The Cars	4 wks	2
5.	The DJ	1 wk	2
6.	The Dress - Find the Perfect Dress	3 wks	2
7.	The Venue - Confirm Menu	4 wks	2
8.	The Venue - Pay Deposit	2 hrs	7
9.	The Dress - Pay for Dress	1 hr	6
10.	Invitations - Confirm Design	2 wks	6
11.	The Flowers	2 wks	6
12.	Invitations - Confirm Numbers	1 wk	10
13.	Invitations - Invites Produced	2 mons	12
14.	Review progress	4 wks	3,4,5,8,9,11,12,13
15.	Invitations - Send Out	1 wk	13
16.	The Dress - First Fitting	2 hrs	9
17.	Invitations - Review RSVP's	1 wk	15
18.	The Venue - Confirm Numbers	2 hrs	17
19.	The Venue - Confirm Timings	2 hrs	18
20.	The Venue - Pay Balance	1 hr	19
21.	The Dress- Final Fitting	2hr	20

- Create a milestone task at the end: The Big day (P) (Dependencies: 19,21)
- Creating the summary tasks :
 - Before 6 month to go (Task 2,3,4,5,6,7)
 - 6 month to go (Task 8,9,10,11)
 - 5 month to go (Task 12,13)
 - 4 month to go (Task 14)
 - 3 month to go (Task 15)
 - 2 month to go (Task 16,17)
 - 1 month to go (Task 18,19,20,21)
- Establish the resource table:

STT	Name	Type	Initials	Groups	Max. Units	Std. Rate
-----	------	------	----------	--------	------------	-----------

1.	Venue	Material	Initials			£3,000.00
2.	Photographs	Material	V			£1,000.00
3.	Flowers	Material	P			£900.00
4.	Cars	Material	F			£500.00
5.	DJ	Material	C			£250.00
6.	Guest	Work	D		8000%	£25/1hr
7.	Cost					

- Assign the resource for each task in the project

STT	Name	Resource
1.	The Engagement	
2.	The Venue - Confirm Dates	Venue[1]
3.	The Photographer	Photographs[3]
4.	The Cars	Cars[1]
5.	The DJ	DJ[1]
6.	The Dress - Find the Perfect Dress	Cost [10]
7.	The Venue - Confirm Menu	
8.	The Venue - Pay Deposit	Cost[£120]
9.	The Dress - Pay for Dress	Cost [£210]
10.	Invitations - Confirm Design	Cost[£90]
11.	The Flowers	Flowers[1]
12.	Invitations - Confirm Numbers	
13.	Invitations - Invites Produced	
14.	Review progress	
15.	Invitations - Send Out	
16.	The Dress - First Fitting	
17.	Invitations - Review RSVP's	
18.	The Venue - Confirm Numbers	
19.	The Venue - Confirm Timings	
20.	The Venue - Pay Balance	Guests[7,000%]
21.	The Dress- Final Fitting	
22.	The Dress- Final Fitting	

Capture images:

- Gantt Chart (Resource Name, Cost Column)
- Display the Project Information (Statics): Duration, Hours, Cost
- Resource Sheet
- Resource Usage

No.	Name	Type	Quantity	Unit	Std.Rate	Base	Related Tasks
1	Project Manager	Work	1	Person	20	PM_Time	Project Planning
2	System Analyst	Work	3	Person	15	Extra_Time	Requirement Analysis, Database Design
3	Backend Developer	Work	4	Person	18	Dev_Time	Backend development, Frontend & Backend Integration
4	Frontend Developer	Work	3	Person	17	Dev_Time	Frontend development
5	UI/UX Designer	Work	2	Person	16	Extra_Time	UI/UX design
6	Tester	Work	3	Person	16	Dev_Time	Testing & bug fixing
7	Cloud Server (AWS, Azure)	Cost	1	Subscription	200		Data storage, application deployment
8	Software Licenses	Cost	1	Software Package	99		Deployment to Test Environment
9	Developer Workstations	Material	4	Units	1200		Deployment to Test Environment
10	UI/UX Design Tools	Cost	1	Software Package	200		Official Deployment
11	Testing Tools (Selenium, JMeter)	Cost	1	Software Package	90		Testing & bug fixing, Evaluation & Optimization

Capture images:

- Gantt Chart (Resource Name, Cost Column)
- Display the Project Information (Statics): Duration, Hours, Cost
- Resource Sheet
- Resource Usage

Exercise 6. The Engagement planning

Creating the following tasks

STT	Name	Duration	Dependencies
1.	The Engagement	1 day	
2.	The Venue - Confirm Dates	4 wks	1
3.	The Photographer	3 wks	2
4.	The Cars	4 wks	2
5.	The DJ	1 wk	2
6.	The Dress - Find the Perfect Dress	3 wks	2
7.	The Venue - Confirm Menu	4 wks	2
8.	The Venue - Pay Deposit	2 hrs	7
9.	The Dress - Pay for Dress	1 hr	6
10.	Invitations - Confirm Design	2 wks	6
11.	The Flowers	2 wks	6
12.	Invitations - Confirm Numbers	1 wk	10
13.	Invitations - Invites Produced	2 mons	12
14.	Review progress	4 wks	3,4,5,8,9,11,12,13
15.	Invitations - Send Out	1 wk	13
16.	The Dress - First Fitting	2 hrs	9
17.	Invitations - Review RSVP's	1 wk	15
18.	The Venue - Confirm Numbers	2 hrs	17
19.	The Venue - Confirm Timings	2 hrs	18
20.	The Venue - Pay Balance	1 hr	19
21.	The Dress- Final Fitting	2hr	20

- Create a milestone task at the end: The Big day (P) (Dependencies: 19,21)
- Creating the summary tasks :
 - Before 6 month to go (Task 2,3,4,5,6,7)
 - 6 month to go (Task 8,9,10,11)
 - 5 month to go (Task 12,13)
 - 4 month to go (Task 14)
 - 3 month to go (Task 15)
 - 2 month to go (Task 16,17)
 - 1 month to go (Task 18,19,20,21)
- Establish the resource table:

STT	Name	Type	Initials	Groups	Max. Units	Std. Rate
-----	------	------	----------	--------	------------	-----------

1.	Venue	Material	Initials			£3,000.00
2.	Photographs	Material	V			£1,000.00
3.	Flowers	Material	P			£900.00
4.	Cars	Material	F			£500.00
5.	DJ	Material	C			£250.00
6.	Guest	Work	D		8000%	£25/1hr
7.	Cost					

- Assign the resource for each task in the project

STT	Name	Resource
1.	The Engagement	
2.	The Venue - Confirm Dates	Venue[1]
3.	The Photographer	Photographs[3]
4.	The Cars	Cars[1]
5.	The DJ	DJ[1]
6.	The Dress - Find the Perfect Dress	Cost [10]
7.	The Venue - Confirm Menu	
8.	The Venue - Pay Deposit	Cost[£120]
9.	The Dress - Pay for Dress	Cost [£210]
10.	Invitations - Confirm Design	Cost[£90]
11.	The Flowers	Flowers[1]
12.	Invitations - Confirm Numbers	
13.	Invitations - Invites Produced	
14.	Review progress	
15.	Invitations - Send Out	
16.	The Dress - First Fitting	
17.	Invitations - Review RSVP's	
18.	The Venue - Confirm Numbers	
19.	The Venue - Confirm Timings	
20.	The Venue - Pay Balance	Guests[7,000%]
21.	The Dress- Final Fitting	
22.	The Dress- Final Fitting	

Capture images:

- Gantt Chart (Resource Name, Cost Column)
- Display the Project Information (Statics): Duration, Hours, Cost
- Resource Sheet
- Resource Usage

MODULE 5: COST, CONSTRAINS, DEADLINE for TASKs

Exercise 1. Entering Cost

- Open the Exercise 1 (Module 4) [Cost: \$1.984]
- Tab View → Resource Sheet
- Add the resource for this project bellow: Type Cost (Not set value, Rate)

STT	Name	Type
11.	Cost Tax	Cost
12.	Logistic Cost	Cost

Assign Cost for task **Ready for preproduction task (Cost Task: 10\$, Logistic Cost: 10\$)**

- Cost of project?
- Update Project after 5 days? Cost of Project?

Exercise 2.

- Exercise 2 (Module 4) Cost: ?
- Tab View → Resource Sheet
- Add the resource for this project bellow: Type Cost (Not set value, Rate)

STT	Name	Type
5.	Cost Tax	Cost
6.	Logistic Cost	Cost
7.	OT Cost	Cost
8.	Bonus	Cost
9.	Punishment	Cost

- Assign Cost for these tasks:

ID	TASK NAME	RESOURCE
1	Visit estate agents	MDF [2], OT Cost[50]
2	View properties	Flooring [2], Cost Task[3]
3	Decide what to buy	Purple Material [1], Bonus[12]
4	Make offer	Flooring [1], Purple Material [1], OT Cost[8], Punish[5]
5	Wait for decision	Flooring [1], Orange Material [1], Logistic Cost[75]
6	offer a accepted	Orange Material [2], Cost Tax [4], Bonus[10]

- Cost of project?
- Update Project after 3 days? Cost of Project?

Exercise 3.

- Exercise 3 (Module 4) Cost: ?
- Tab View → Resource Sheet
- Add the resource for this project bellow: Type Cost (Not set value, Rate)

STT	Name	Type
-----	------	------

10.	Tax	Cost
11.	Upgrade Fee	Cost
12.	Maintain Fee	Cost
13.	Bonus	Cost
14.	Part Time Fee	Cost

- Assign the resource for each task

ID	TASK NAME	Resource
1	Planning	PM, Tax[4]
2	Requirement documents	PM, Designer, Part Time Fee[10]
3	Sign off	PM
4	Development	Developer, Operating System[1] , Visual Studio [1], Upgrade Fee[21]
5	Testing	Tester, Part Time Fee[10], Bonus[18]
6	Implementation	Maintainer, Operating System , Visual Studio, Cable [100], Maintain Fee[18], Bonus[23]
7	Review	PM, Maintainer

- Cost of project?
- Update Project after 2 weeks? Cost of Project?

Exercise 4.

- Exercise 4 (Module 4) Cost: ?
- Tab View → Resource Sheet
- Add the resource for this project bellow: Type Cost (Not set value, Rate)

STT	Name	Type
26.	Tax	Cost
27.	Bonus	Cost
28.	Part Time Fee	Cost
29.	Logictis Cost	Cost

- Assign the resource for each task

ID	TASK NAME	RESOURCE
5	Planning Completed	Part Time Fee[150], Bonus[70]
12	Site works Completed	Logictis Cost[240]
20	Building Construction Completed	Logictis Cost[1004], Bonus[210]
33	Fit out Completed	Tax[50]
42	Commissioning Completed	Bonus[180]

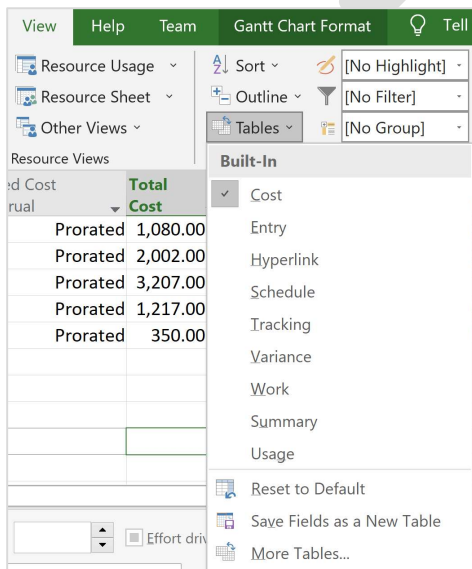
- Cost of project?
- Update Project after 1 month? Cost of Project?
- Update Project after 4,5 month? Cost of Project?
- Update Project after 6 month? Cost of Project?

MODULE 6: Hiện thị dự án theo một số yêu cầu- Report

- MS Project cho phép hiển thị các task trong dự với các cách khác nhau như:

Calendar	Lịch hàng tháng chỉ ra các công việc và khả năng hoàn thành nó.
Gantt Chart	Diễn tả các công việc và các thông tin có liên quan, một biểu đồ (biểu đồ ngang) thể hiện các công việc và thời gian hoàn thành chúng.
Network Diagram	Thể hiện dưới dạng lưới các công tác (sơ đồ mạng) và sự phụ thuộc giữa chúng. Dùng khung nhìn này để có một cái nhìn bao quát về các công việc.
Task Usage	Thể hiện danh sách các công việc đã được gán tài nguyên. Dùng khung nhìn này để thấy nguồn nguồn lực nào đã được gán cho một công việc cụ thể
Tracking Gantt	Thể hiện danh sách của công việc và thông tin có liên quan. Dùng khung nhìn này để theo dõi tiến trình của dự án.
Resource Graph	Thể hiện biểu đồ phân phối tài nguyên. Dùng khung nhìn này để thể hiện thông tin về một nguồn lực dưới các tiêu chí khác nhau.
Resource Sheet	Danh sách nguồn lực và thông tin liên quan. Dùng khung nhìn này để nhập và hiệu chỉnh các thông tin về tài nguyên.

- Tab View: Hiện thị và cho phép lọc theo các tham số khác nhau



Thể hiện các giá trị cần lọc

Hiện thị dự án theo chi phí

Hiện thị dự án toàn bộ dự án

Hiện thị dự án theo lịch thực hiện

Theo dõi toàn dự án

Hiện thị dự án theo tổng các work thực hiện

- So sánh các giá trị Cost, Baseline, Variance, Actual

Work: Thời gian thực hiện theo các tham số hiện tại.

Baseline: Thời gian thực hiện công việc theo kế hoạch.

Variance: Khoảng thời gian thay đổi giữa thực tế và kế hoạch.

Actual: Thời gian thực tế đã thực hiện được theo các tham số hiện tại.

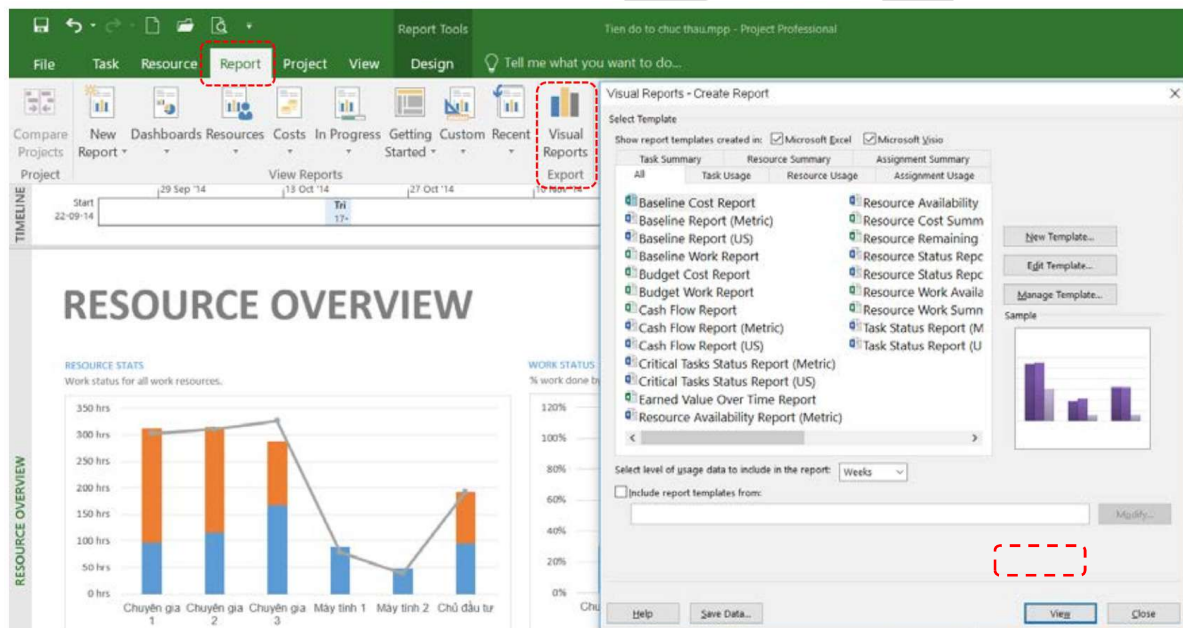
Remaining: Thời gian thực tế còn lại.

%W. Coml: Số phần trăm công việc đã thực hiện.

- Xuất các báo cáo (Report) cho dự án

1. Báo cáo theo đồ thị:

Tab Report → Visual Reports



2. Báo cáo cơ bản dạng số liệu

Tab Report → View Group

Resource	Report	Project	View	Help	Team	Gantt Chart Format	Tell me what you want to do
Dashboards	Resources	Costs	In Progress	Getting Started	Custom Recent	Visual Reports	Export
Task Name	Start	Finish	Baseline Start	Baseline Finish	Start Var.	Finish Var.	Add New
Thu thập yêu cầu	18/10/2021	25/10/2021	NA	NA	0 days	0 days	
Phân tích thiết kế	26/10/2021	16/11/2021	NA	NA	0 days	0 days	
Viết chương trình	05/11/2021	25/11/2021	NA	NA	0 days	0 days	
Kiểm thử	26/11/2021	03/12/2021	NA	NA	0 days	0 days	
Triển khai	06/12/2021	06/12/2021	NA	NA	0 days	0 days	

Task Name	Duration	
Khảo sát	2 days	Bùi Hoàng Quyên, Thuế[20.00 đ]
Phân tích	4 days	Framework[1], Lê Lan Phương[200%], Chi phí đi lại[5.00 đ]
Thiết kế	7 days	OS[2], Trần Ngọc Hà
Xây dựng	0.5 mons	Trần Ngọc Hà, Hardisk[0.5]
Kiểm thử	1 wk	Hoàng Mạnh Quân (PM), Thuế[10.00 đ]
Chuyển Giao	4 days	Lê Hoàng Minh (TV)[50%], Mai Hà Linh, Chi phí công tác[100.00 đ]
Hướng dẫn sử dụng	3 days	Phạm Minh Vương (TV), Chi phí đi lại[10.00 đ]

Resource Name	Type	Material Label	Initials	Group	Max. Units	Std. Rate	Ovt. Rate	Cost/Use
Hoàng Mạnh Quân (PM)	Work		H		100%	10.00 đ/hr	0.00 đ/hr	20.00 đ
Lê Hoàng Minh (TV)	Work		L		50%	20.00 đ/hr	0.00 đ/hr	10.00 đ
Phạm Minh Vương (TV)	Work		P		200%	15.00 đ/hr	0.00 đ/hr	30.00 đ
Lê Lan Phương	Work		L		100%	15.00 đ/hr	0.00 đ/hr	10.00 đ
Hoàng Lê Minh	Work		H		100%	10.00 đ/hr	0.00 đ/hr	0.00 đ
Mai Hà Linh	Work		M		100%	15.00 đ/hr	0.00 đ/hr	10.00 đ
Trần Ngọc Hà	Work		T		200%	15.00 đ/hr	0.00 đ/hr	10.00 đ
Bùi Hoàng Quyên	Work		B		100%	15.00 đ/hr	0.00 đ/hr	10.00 đ
OS	Material		O			250.00 đ		0.00 đ
Hardisk	Material		H			120.00 đ		0.00 đ
USB	Material		U			1.00 đ		0.00 đ
Framework	Material		F			80.00 đ		2.00 đ
Chi phí đi lại	Cost		C					
Chi phí công tác	Cost		C					
Thuế	Cost		T					